

Decision-State Diagnostics for Healthcare AI

Why Agentic Clinical Systems Need More Than Confidence Scores

Healthcare AI is moving quickly from passive summarization toward systems that retrieve evidence, extract structured data, triage messages, recommend next steps, and coordinate work across clinical and administrative settings. That shift changes the uncertainty problem. It is no longer enough to ask whether a model is confident in a final answer. A healthcare workflow often needs to ask what kind of decision state the system is in.

Is the next action stable enough to proceed? Is the uncertainty concentrated between two plausible choices? Is it diffuse across many alternatives? Is the system sensitive to a boundary condition? Has the decision state shifted over a multi-step trajectory?

Decision-PGA is a prototype framework for studying those questions. It analyzes clouds of categorical probability vectors on the probability simplex using Fisher-Rao/square-root geometry, then maps the resulting dispersion shape into states such as stable, binary ambiguity, diffuse uncertainty, boundary sensitivity, and regime shift. In plain language: it tries to describe the shape of uncertainty around a decision, not merely the size of that uncertainty.

This article is a personal technical perspective, not an institutional statement. It is motivated by high-accountability healthcare environments, including academic medical centers such as Mayo Clinic, where AI is actively being explored for clinical, operational, and research workflows. It does not represent Mayo Clinic policy or endorsement. The work described here uses no patient data, is not clinical validation, and is not a medical device or clinical decision support product.

Why healthcare is a natural stress test

Healthcare is full of decisions that are too consequential for a single opaque confidence score:

- A patient-message assistant must decide whether to answer, clarify, route to a nurse, route to a physician, retrieve chart context, or escalate.
- A guideline or policy assistant must decide whether the retrieved evidence is coherent, stale, conflicting, or insufficient.
- A document-extraction system must decide whether a medication dose, prior

- A coding or revenue-cycle assistant must decide whether extracted evidence supports a code, conflicts with another source, or needs human review.
- A care-management agent must decide whether to proceed with a plan step, gather more evidence, ask a clarifying question, or replan.

These are not all clinical diagnosis tasks. Many are operational or administrative. But they share a common structure: a system is choosing among candidate actions, labels, sources, or extracted values under uncertainty. That is exactly the space where a decision-state diagnostic can be useful.

The regulatory and governance backdrop

Healthcare AI is not an abstract future category. The FDA maintains a public resource of AI-enabled medical devices authorized for marketing in the United States:

<https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-and-machine-learning-aiml-enabled-medical-devices>.

The FDA also has guidance on Clinical Decision Support software and how some software functions fit within or outside device regulation:

<https://www.fda.gov/medical-devices/software-medical-device-samd/clinical-decision-support-software-frequently-asked-questions-faqs>.

The ONC HTI-1 final rule adds transparency requirements for predictive decision support interventions in certified health IT:

<https://healthit.gov/regulations/hti-rules/hti-1-final-rule/>. The WHO has published guidance on ethics and governance of AI for health: <https://www.who.int/publications/i/item/9789240029200>.

Healthcare institutions are also visibly building AI capacity. Public Mayo Clinic pages describe AI as a major area of medical innovation:

<https://www.mayoclinic.org/giving-to-mayo-clinic/our-priorities/artificial-intelligence>.

The Mayo Clinic Platform page describes work around secure, de-identified clinical data and digital health innovation:

<https://www.mayoclinic.org/giving-to-mayo-clinic/our-priorities/mayo-clinic-platform>.

These sources point in the same direction: healthcare AI needs transparency, validation, governance, and practical control surfaces. Decision-PGA is not a regulatory framework. But it aims at a technical gap underneath many governance questions: how can an AI workflow expose the state of its own decisions in a readable, testable way?

Confidence is not enough

Suppose a system must choose between several next actions:

- retrieve more evidence;
- route to a clinician;
- abstain;
- replan.

Two cases can have similar entropy but require different responses. In one case, almost all uncertainty may lie between two options, such as answer now versus ask one specific clarifying question. In another, the probability mass may be scattered across many actions, suggesting that the system does not have enough context to know what kind of move is appropriate.

A scalar confidence score may say "uncertain" in both cases. A decision-state diagnostic tries to say more:

- tight cloud: proceed may be reasonable;
- elongated two-label cloud: clarify between top options;
- diffuse cloud: gather more evidence;
- boundary-sensitive cloud: inspect assumptions or sensitivity;
- drifting cloud: segment the task or replan.

This distinction matters in healthcare because the next action is part of the safety case. The same uncertainty score should not always produce the same workflow response.

What Decision-PGA does

Decision-PGA starts with repeated probability-like observations over candidate labels. Those labels might be actions, extracted field values, evidence clusters, review outcomes, or routing decisions. The current prototype then:

1. Normalizes probability vectors.
2. Maps them to the positive unit sphere with the square-root transform.
3. Computes an intrinsic mean.
4. Log-maps samples into a tangent space.
5. Computes a dispersion tensor and eigensystem.
6. Reports shape metrics such as total dispersion, PC1 fraction, anisotropy ratio, margin, label-switching, and half-window geodesic drift.

The practical output is not just a metric table. It is an action-oriented diagnostic contract:

State Possible workflow interpretation
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Stable Proceed, assuming the task is within scope.
Binary ambiguity Ask a targeted question or compare the top candidates.

This is still a research scaffold. It must be tested against baselines such as entropy, margin, switch rate, agreement, and task-specific accuracy. But the core idea is simple: if healthcare AI systems are going to act in workflows, they need diagnostics for the state of their decisions.

Healthcare use cases worth studying first

Patient-message and inbox triage

Many healthcare workflows begin with a message. The candidate decisions may be answer, ask for clarification, route to nurse, route to physician, schedule, retrieve chart context, or escalate. Decision-PGA could help distinguish a stable routing decision from a two-route ambiguity or diffuse uncertainty.

Guideline and policy retrieval

RAG-style systems can retrieve evidence that is relevant but not coherent. The system may face conflicts between guidelines, local policies, patient-specific constraints, or outdated sources. A diagnostic over candidate evidence clusters could help decide whether to answer, retrieve more, cite uncertainty, or route to human review.

Medical document extraction

Healthcare uses enormous volumes of forms, referrals, notes, faxes, PDFs, prior authorizations, medication lists, and discharge documents. Extraction uncertainty is often not just "low confidence." It may be a two-value dispute, a source-span ambiguity, a table-row association problem, or OCR sensitivity. Decision-PGA could sit downstream of extraction candidates and help triage the kind of review needed.

Clinical trial matching

Eligibility matching requires structured decisions over inclusion and exclusion criteria. Some fields are stable; some are missing; some conflict across notes or reports. Decision-state diagnostics could help separate clear matches from missing-evidence cases and true ambiguity.

Care-management and operational agents

Agentic systems may coordinate multi-step workflows: gather context, check criteria, prepare documentation, route work, and monitor follow-up. A final confidence score misses drift across the trajectory. Sliding-window decision-state diagnostics could help identify when the agent's plan has moved into a different regime.

Decision-PGA is not clinical validation. It is not a substitute for model evaluation, usability testing, bias assessment, safety review, regulatory analysis, or clinical governance. It does not determine truth. It does not decide whether a diagnosis, treatment, or recommendation is correct. It only analyzes the shape of probability-like observations over candidate decisions.

That limitation is important. In healthcare, a tool that describes uncertainty can still be useful, but only if it is surrounded by validation, human oversight, workflow design, and governance appropriate to the use case.

The prototype should therefore be evaluated first on synthetic and retrospective fixtures:

- controlled action-selection examples;
- document-extraction candidate fixtures;
- evidence-conflict examples;
- simulated agent trajectories;
- benchmark comparisons against entropy and margin.

Only after those tests should anyone consider prospective workflow studies, and only with the appropriate institutional, regulatory, privacy, and ethics review.

Why publish now

The timing is useful because the healthcare AI conversation is shifting from "can models answer questions?" to "how do AI systems participate in workflows responsibly?" That shift requires technical vocabulary for states between confidence and failure.

Decision-PGA gives one possible vocabulary:

- stable decisions;
- structured ambiguity;
- diffuse uncertainty;
- sensitivity near a boundary;
- decision-state drift.

It is not the final answer. It is a proposal for a measurement layer that can be made explicit, tested, criticized, and improved.

A near-term open-source path

The most practical next step is not to overclaim the method. It is to publish a clear article, release the prototype openly, and invite technical feedback.

- reproducible synthetic benchmarks;
- document-extraction and healthcare-flavored examples that use no patient data;
- local CLI and MCP-style tool interfaces;
- readable Markdown/PDF reports;
- clear limitations.

The strongest early healthcare-facing benchmark would not require clinical data. It could use synthetic fixtures that mimic common workflow decisions: two plausible extracted values, conflicting evidence clusters, routing ambiguity, and multi-step drift.

Conclusion

Healthcare AI will need more than confident answers. It will need systems that can expose when they are stable, ambiguous, diffuse, sensitive, or drifting.

Decision-PGA is a small attempt to formalize that layer. By analyzing probability clouds on the simplex, it asks whether the geometry of a decision state can help route AI workflows toward better next actions: proceed, clarify, gather evidence, inspect sensitivity, segment, replan, or escalate.

That question is worth studying now, before agentic healthcare systems become routine infrastructure. The right goal is not to claim that Decision-PGA is ready for clinical deployment. The right goal is to make decision-state diagnostics visible enough that they can be tested before they are needed at scale.